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Integrating Federated Learning and Explainable AI for Plant Disease Detection: A Systematic Review of Recent Advances in Precision Agriculture

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Abstract

Early detection of plant and crop diseases is vital for achieving sustainable agriculture and global food security. Traditional inspection methods are often slow and subjective, whereas Artificial Intelligence (AI) techniques offer fast, scalable, and objective alternatives. Following the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines, this systematic review synthesizes 145 studies published between 2023 and 2025 that employ Machine Learning (ML), Deep Learning (DL), Explainable AI (XAI), and Federated Learning (FL) for plant and crop disease classification. The studies are organized by crop species, imaging modality, and model architecture to evaluate performance in terms of accuracy, robustness, interpretability, and privacy preservation. Results reveal that XAI techniques such as Grad-CAM, LIME, and SHAP enhance transparency and trust, while FL enables decentralized and privacy-aware collaboration across distributed agricultural datasets with only minimal reductions in model accuracy compared to centralized training. Despite strong results in controlled conditions, many models struggle to generalize under real-field variability due to data imbalance and environmental factors. Emerging directions include lightweight edge architectures, domain adaptation, and unified explainable-federated frameworks. Overall, this review identifies FL and XAI as complementary drivers of transparent, privacy-preserving, and scalable AI systems for sustainable precision agriculture.

Keywords Disease Detection · Federated Learning · Explainable AI · Precision Agriculture

1 Introduction

Timely identification of plant and crop diseases is essential to protecting global agriculture, as infections caused by fungal, bacterial, or viral pathogens can substantially impair plant metabolism and result in significant yield and quality losses [1–5]. According to the FAO (Food and Agriculture Organization), plant and crop diseases cause 20–40% of annual global agricultural losses representing a significant barrier to sustainable production of crops and the achievement of global food securities [6, 7]. Such losses strain the global food systems, and they



exacerbate socio-economic vulnerabilities particularly in agriculture dependent economies. Resulting in lower farm income, more volatility in the prices of the market and ultimately less growth economy at a national level and therefore must guide themselves on disease management. In precision agriculture, timely and accurate diagnosis of plant diseases is imperative for increasing production and reducing losses [8]. In this context, recent advances in AI-enabled smart farming and IoT-integrated agricultural systems have demonstrated the value of combining intelligent analytics with field-level sensing, including smart cultivation frameworks, solar-powered multi-layer soil moisture monitoring, and AI-based soil moisture measurement and prediction [9–12]. Such integrated sensing and data-driven approaches provide early indicators of plant stress and environmental conditions, thereby complementing disease surveillance and reinforcing the need for automated, scalable diagnostic solutions. Manual disease detection methods are slow, labor-heavy, and subjective, making them unfit for large-scale farming due to reliance on expertise, environment, and symptom stage [13]. By the time visual symptoms appear, the infection may have already propagate across the field, causing irreversible harm to crop health and yield potential [14].

Building upon this issue, ML and DL techniques have transformed plant disease detection by identifying complex patterns in large datasets that human inspection could readily miss [15–18]. These models analyze leaf images rapidly and accurately, surpassing conventional methodologies in consistency and scalability [19]. The deployment of intelligent tools on precision agriculture allows for monitoring, executing targeted interventions and sustainable farming, accordingly improving resilience and global food security [20]. ML and DL based computational tools have revolutionized agriculture by providing automated early disease detection cost effectively at scale using image analysis from RGB, hyperspectral, and multispectral sources replacing manual inspection [13]. By leveraging image data from RGB, hyperspectral, and multispectral sensors, AI-driven methods offer cost-effective and automated diagnosis that mitigates chemical usage and reduces crop losses [21].

Nevertheless, despite these advancements, practical limitations persist. Many existing models are trained on limited or unbalanced datasets that lack diversity in crop species, disease stages, and regional representation, resulting in overfitting and poor generalization in real-world conditions [22]. Variations in lighting, weather, and background complexity further hinder model robustness and reliability in non-standardized settings. Moreover, privacy and data ownership concerns limit large-scale data sharing across farms and institutions. To address these issues, state-of-the-art paradigms such as FL and XAI have emerged as powerful extensions to traditional ML/DL frameworks, enabling privacy-preserving collaboration and interpretable decision-making [23].

FL constitutes a new frontier in agricultural AI, enabling decentralized and privacy-aware model training without the need to transmit raw data among stakeholders [24, 25]. With the rapid growth of IoT sensors, UAV imagery, and mobile data collection, agricultural information has become extensively dispersed across regions and institutions. FL allows farms, research centers, and cooperatives to collaboratively train shared models while retaining data locally, thereby preserving data sovereignty and minimizing communication overhead [26, 27]. This decentralized approach ensures trust among participants, mitigates data heterogeneity, and fosters scalable learning across diverse agro-ecological zones characterized by differences in soil, climate, and disease patterns [28]. In recent applications, FL has been incorporated into crop disease detection pipeline, where farms are capable of detect and classify the diseases in crops like papaya, potato, tomato, banana avocado and sunflower jointly [29–34]. As agriculture faces escalating challenges from climate change, population growth, and food insecurity, FL offers a transformative pathway for integrating distributed intelligence into sustainable and responsive crop management systems [35].

While FL primarily emphasizes privacy and scalability, another critical dimension in agricultural AI is model interpretability. Many high-performing DL models function as “black boxes,” producing accurate predictions without providing interpretative understanding of their reasoning processes [36]. This opacity limits trust, especially in field applications where understanding why a model made a decision is essential for informed intervention and farmer confidence [37]. XAI addresses this limitation by making model predictions transparent and interpretable through visualization and feature attribution techniques such as Gradient-weighted Class Activation Mapping (Grad-CAM), Local Interpretable Model-agnostic Explanations (LIME), and SHapley Additive

exPlanations (SHAP) [38]. In agriculture, XAI for disease detection can facilitate the identification of which visual or other environmental characteristics most contribute to the categorization of unhealthy versus healthy plants and minimize false-positives. For instance, visualization tool such as the class activation maps have been employed on their mobile application with real-time plant disease identification in which farmers are able to follow back model predictions up to the leaf regions to enable localized interpretability [39]. Hybrid models that combine classical ML with interpretable features is known to be transparent and computationally effective models in the resource-constrained agricultural setup [40]. Moreover, the fusion of XAI with monitoring systems based on IoT enables real-time diagnosis and decision-making in intelligent farming. Through providing improved accuracy and interpretability of the AI systems, XAI serves a pivotal role in creating trust among farmers so that decisions taken can be actionable, context-aware, and aligned with sustainable agricultural practices [41].

Given these developments, there is a growing need to combine interpretability and privacy preservation in intelligent agricultural systems. The convergence of FL and XAI represents a promising paradigm that bridges transparency, trust, and secure collaboration, the three pillars essential for the future of precision agriculture. The present study is motivated by this convergence and aims to synthesize recent advancements while identifying key research gaps and opportunities for improvement.

1.1 Motivation and Research Contributions

Motivated by the escalating demand for transparent and privacy-preserving AI solutions, this review is motivated by the need to improve early disease detection in plants and crops through federated and explainable learning frameworks. Conventional diagnostic methodologies are time-consuming, subjective, and unsuitable for large-scale agricultural monitoring. Although data-driven artificial intelligence (AI) methods especially ML and DL have shown impressive results in identifying diseases from image data, multiple challenges persist. These include limited data diversity, poor model generalization to real-field conditions, lack of interpretability, and growing concerns over data privacy and ownership.

Agricultural data are inherently decentralized, collected from farms, laboratories, and regional institutions that operate under varying environmental and technological conditions. Centralizing such sensitive and geographically distributed datasets is often impractical and raises privacy challenges. FL effectively mitigates these challenges by enabling multiple stakeholders to collaboratively train a shared global model without necessitating the exchange of raw data, thereby preserving data sovereignty and privacy. FL also mitigates non-Independent and Identically Distributed (non-IID) data heterogeneity while improving scalability and cross-domain adaptability.

In parallel, XAI enhances the transparency and trustworthiness of these intelligent computational systems by revealing the reasoning behind model predictions through interpretability techniques including Grad-CAM, LIME, and SHAP. When combined, FL and XAI form a complementary paradigm linking privacy-preserving collaboration with interpretability to advance reliable and ethically responsible precision-agriculture solutions. While several recent surveys have addressed DL techniques for plant disease detection, this review differs fundamentally in both scope and analytical focus.

While numerous survey articles have examined ML and DL approaches for plant disease detection, existing reviews predominantly focus on centralized learning paradigms and performance-driven model comparisons. In contrast, the key novelty of this review lies in its integrated and systematic examination of both FL and XAI techniques within the plant disease detection domain. By jointly analyzing FL and XAI, this review addresses two critical and complementary requirements for real-world agricultural deployment: data privacy preservation across distributed farms and institutions, and model transparency and trustworthiness for end users such as farmers and agronomists.

This dual-perspective analysis enables insights into deployment feasibility, robustness, and user trust that are not captured in existing surveys, thereby positioning this review as a distinct and timely contribution to the literature.

1.2 Research Questions

In this review, we present the recent progress in this field and discuss their advantages and limitations as well as future research directions for disease detection methods. This review articulates key research inquiries and outlines its main contribution areas below:

- **RQ1:** What are the major plant species and corresponding diseases investigated in the literature, and how accessible and generalized are the datasets used for plant disease detection?
- **RQ2:** How does different environmental factors impact the robustness and generalization ability of plant disease detection models?
- **RQ3:** Which ML, DL, and hybrid ML–DL models are employed for plant disease detection, and how do they compare with respect to performance, computational efficiency, and scalability?
- **RQ4:** What is the impact of using XAI, and FL methods on improving transparency, privacy preservation and collaborative intelligence in plant and crop disease detection systems?
- **RQ5:** What are challenges and future directions for the development of scalable, interpretable, environmentally robust AI systems for plant and crop disease detection beyond state-of-the-art?

2 Research Methodology and Review Protocol

This study adopts a comprehensive systematic review methodology to examine existing techniques for the early identification of plant and crop diseases. The review encompasses ML, DL, and hybrid frameworks, along with emergent paradigms including XAI and FL. These approaches are analyzed from multiple perspectives, including robustness, scalability, interpretability, and privacy-preserving collaboration. The reviewed studies are systematically organized according to crop and plant types, disease categories, imaging modalities, and model performance metrics. All selected articles were critically screened, analyzed, and categorized to ensure a coherent and methodologically structured synthesis of state-of-the-art advances in intelligent plant disease detection.

2.1 Screening Criteria and Inclusion Procedures

The screening process allowed us to include studies on crop and plant disease detection based predominantly on image-based ML, DL, hybrid, FL and XAI techniques employing computational architectures such as CNNs, YOLO, Transfer Learning and Transformer models. The identified literature was subjected to a manual screening process comprising three distinct phases:

- **Title screening:** Documents were rejected if their titles did not correspond with the topic.
- **Abstract screening:** Papers were eliminated if their abstracts failed to meet any of the inclusion criteria.
- **Full-text screening:** Documents were excluded if their complete content failed to satisfy all inclusion criteria.

2.2 Inclusion Criteria

Articles were considered eligible for inclusion if they met the following conditions. First, the study addressed plant or crop disease analysis, including disease detection, identification, localization, or severity estimation. Second, the proposed approach was based on ML or DL techniques, with explicit incorporation of FL, explainable artificial intelligence, or a combination of both. Third, the work reported an original methodological or experimental contribution, supported by empirical evaluation. Fourth, adequate technical information was provided, including dataset characteristics, experimental configuration, and quantitative performance metrics. Finally, only peer-reviewed publications written in English and published between 2023 and 2025 were included to reflect

recent developments. These criteria were applied systematically to ensure consistency, relevance, and reproducibility in line with PRISMA guidelines.

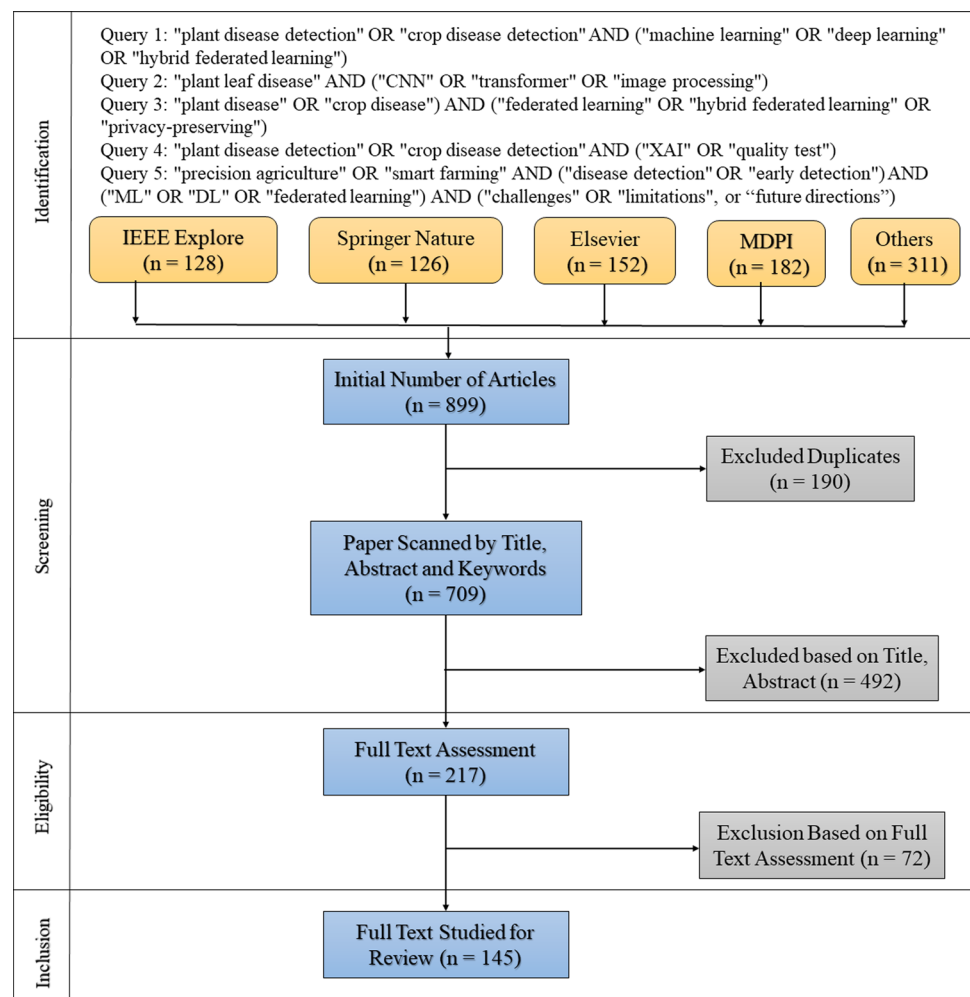
2.3 Strategy, Study Selection, and Data Extraction

This research reviews systematically reviews existing research efforts on the integration of ML, DL, and FL methodologies in plant and crop disease detection. To retrieve pertinent scholarly literature, we proposed an in-depth search strategy consisting of five modified searches which include search keywords such as “plant disease detection,” “CNN,” “transformer,” “federated learning,” “XAI,” and “precision agriculture.” Broad search terms were used in combination with Boolean operators to increase sensitivity but retain specificity.

A total of 899 articles were gathered from major academic databases such as like MDPI (n=182), Elsevier (n=152), IEEE Xplore (n=128), Springer Nature (n=126) and other sources (n=311) such as Frontiers, arXiv, ACS and more. The PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) flow diagram was employed to illustrate the search and selection process of eligible studies [42, 43].

This procedure is depicted in Fig. 1 in a stepwise manner, such as identification of records, removal of duplicates, title and abstract screening, full text assessment and voluntary study inclusion. After the exclusion of 190 duplicated articles, 709 original articles were screened. A total of 492 papers were excluded due to the irrelevance with title, abstract and key words. The full texts of the remaining 217 studies were retrieved and assessed, and during the full-text screening stage, 72 articles were excluded to ensure methodological rigor, relevance, and

Fig. 1 PRISMA flow diagram for systematic literature review



reproducibility of the review. Studies were removed if they met one or more of the following criteria: (i) the primary focus was not plant or crop disease detection (e.g., yield prediction or general crop monitoring without disease analysis); (ii) the work did not employ ML, DL, explainable AI, or FL techniques; (iii) the study lacked sufficient methodological detail, performance metrics, or experimental validation; (iv) the article was purely conceptual, opinion-based, or survey-like without original technical contribution; (v) the same dataset and model configuration were repeatedly reported with only marginal variations; or (vi) the full text was not accessible or written in a non-English language. These exclusion criteria were applied consistently to improve transparency and reproducibility in accordance with PRISMA guidelines. In addition to the referenced works discussed on PRISMA flow diagram, 43 of the 188 papers reviewed were cited in this article to substantiate the introduction, guide the discourse on motivation and contribution, and underpin the overarching conceptual framework of the study. Finally, a total of 145 articles were included for full review and analysis. Due to heterogeneity in datasets, environmental conditions, and evaluation protocols, direct normalization of performance metrics is not feasible. Therefore, reported accuracy and related metrics are interpreted as within-study indicators, with comparisons focusing on relative trends considering dataset type, model complexity, and deployment context rather than absolute values.

3 Background on AI Techniques for Plant Disease Detection

Plant and crop diseases constitute a major impediment that impedes sustainable agriculture worldwide in terms of productivity and food security. There exists a large number of symptoms, organs and pathogens that have different characteristics making it complex to recognize many diseases in plants and crops. As shown in Fig. 2, these dimensions encompass multiple plant species, varying disease stages, and diverse environmental contexts. This complexity has led to a transition from traditional inspection techniques to AI assisted detecting tool. particularly within image-based diagnostic applications. In the last few years, there has been a large number of studies that utilize ML and DL.

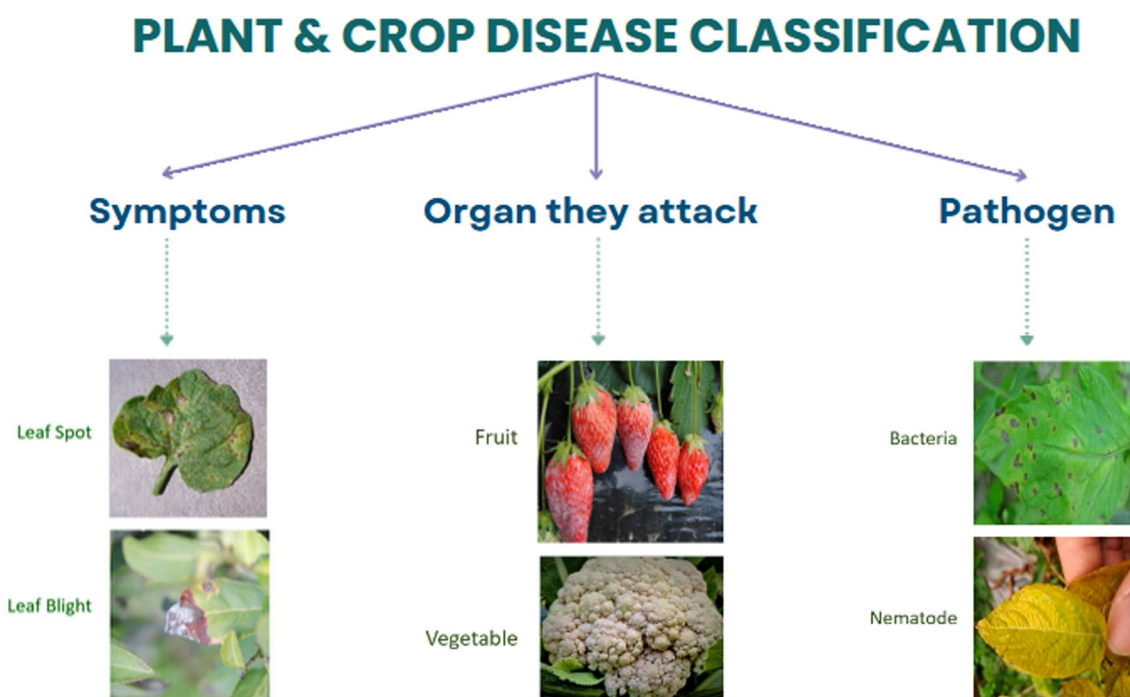


Fig. 2 Plant and crop disease classification

Plant and crop disease classification techniques aiming at detecting crop diseases in plants including tomato, potato, grapevine, and strawberry. Both controlled and real-field datasets have been considered to assess YOLO variants, CNNs, transformers or EfficientNets. Although the performance of these models is high in terms of high accuracy, recall and F1-scores, several critical challenges persist, especially in terms of dataset diversity, real-world generalization and model transparency. Emerging research efforts have increasingly focused on addressing these limitations in XAI for interpretability and FL for collaborative privacy-preserving model training. To address the above challenges, this review makes the following contributions:

- Literature Organization: Clusters studies by disease, crop species and dataset properties, and imaging types.
- State-of-the-art Model Comparison: Contrasts ML, DL and hybrid methods in terms of accuracy, robustness and scalability on various datasets.
- Limitation Analysis: Highlights problems of dataset imbalance, poor environmental diversity and limited model interpretability.
- Advanced Techniques: Reviews the use of Explainable AI and FL for enhancing transparency and data privacy.
- Comparison of Centralize vs Decentralize methods: Compares between centralize (ML, DL and hybrids) and decentralize (FL) methods in terms of accuracy, data privacy and communication cost.

4 Systematic Review of AI-Based Plant Disease Detection Methods

In this section, we examine recent research from 2023 to 2025 that addresses AI-based early detection of crop and plant diseases. The literature emphasizes that although traditional diagnosis is effective in controlled conditions and AI-based approach is scalable and efficient, few studies have been able to combine accuracy with interpretability and robustness under practical agricultural settings. Nevertheless, several critical challenges persist, such as weak generalization, lack of interpretability and lack of robustness when adapting across different crops and disease expressions. An integration with scalable AI designs, model explainability and collaborative learning methods to overcome these challenges can further enable reliable and deployable solutions for contemporary precision agriculture.

To ensure conceptual clarity and avoid overlap in the analysis, the reviewed studies are organized using a two-level categorization strategy. First, studies are grouped according to their core learning paradigm, namely ML, DL, or hybrid ML-DL architectures, based on the primary model used for disease classification. Second, XAI and FL are treated as orthogonal and cross-cutting dimensions that can be integrated with any of these base learning paradigms. Consequently, studies employing XAI or FL are analyzed in dedicated sections to highlight interpretability and privacy-preserving collaboration, even when their underlying models are CNNs, transformers, or hybrid architectures. This structured categorization avoids double counting while enabling a focused and coherent synthesis of methodological trends.

For an extensive review of these aspects, we have organized this section into the following four domains: ML and DL Approaches, hybrid model design approaches, explainable AI techniques and FL mechanisms in detail as follows.

4.1 ML and DL Approaches

Recent developments in ML and DL have made automatic diagnosis of plant diseases an efficient and highly accurate method. Diseases in plants and crops have become a global problem leading to yield losses, while early and accurate detection is necessary for sustainable agriculture. In this subsection, studies are categorized strictly based on their core ML and DL architectures used for plant and crop disease detection. Works that additionally incorporate explainability or federated training mechanisms are discussed here only with respect to their baseline model performance, while their interpretability and privacy aspects are examined separately in Sects. 4.3 and 4.4.

4.1.1 ML Approaches

Classical ML methods for plant disease recognition in plants can generally be delineated into two primary stages: handcrafted features computed from images and followed by classification via conventional algorithmic models. They are typically being used for the interpretability and ease of consideration in contrast to highly computational models. For example, in [44] the authors extracted shape-based features combined with an ANN classifier, which was optimized by GOA for feature selection and produces 98.23% of accuracy. Another study also used Gray-Level Co-occurrence Matrix (GLCM), Local Binary Patterns (LBP) and Hue-Saturation-Value (HSV) histograms as features, then combined with RBF-kernel Support Vector Machine (SVM), which achieved a good accuracy of 91.67% [45]. For example, 1D-ResNets were applied on 56-band hyperspectral reflectance to solve non-RGB data at an accuracy of 91% to 99.02% pointing out the generalization of these methods beyond RGB [46, 47]. For the purpose of cauliflower disease detection, a study combining GLCM and statistical features with a Random Forest classifier [48] was reported as well as in another report similar set of features were used to achieve an accuracy of 82.35% [49]. More advanced work combined K-means clustering with the multiclass SVM using color and texture features [50–52]. A collective drawback of such studies is the use of hand-crafted features, which need manual effort and may not generalize well to challenging real-field conditions with complex backgrounds and illumination changes.

4.1.2 DL Approaches

Customized CNNs serve as the foundation of plant and crop disease classification, it is structures tailored for efficient, low-complexity inference or enhanced feature representation that must model more nuanced field-captured symptoms among tomato, potato, guava, strawberry, and grape [53–58]. In tea, customized deep architectures such as the wavelet-CNNs, gray blight-specific architectures and fusion frameworks have been shown to perform optimal small-target detection and robust multi-scale feature learning in the presence of complex field environments [59–62]. For coffee leaves, pixel-level transfer-feature engineering, parameter-efficient CNNs have been created for joint localization and classification that allow resource-constrained deployments while maintaining strong accuracy [63–68]. Also, in coffee disease studies span CNN-based classification, an enhanced multivariate product representation DeepEMPR detector, digital-agriculture leaf detection and wilt-disease identification using DL [63, 69, 70]. Additionally, a real-time, cost-efficient DL solution for identifying coffee leaf diseases using transfer learning models and in another work a model was developed to classify coffee disease by extracting ROIs, merging overlapping lesions using extended Gaussian KDE [71, 72]. attention-enhanced CNNs also shown good generalization for grape leaf disease discrimination when visual symptoms overlap at testing time [73, 74].

Disease detection and classification systems have been developed via DL pipelines for eggplant and cauliflower by exploiting multi-channel inputs and modern CNN backbones [75–78]. These methods also emphasize the importance of tailored CNN architectures when dealing with particular leaf patterns, texture defects and field-specific conditions typically on field captured datasets. For grape leaves, there is significant efficacy of attention-based CNN models to discriminate diseases having during the testing phase overlapping visual appearances [73]. In eggplant and cauliflower disease classification deep learning-based approaches for classification [76–80].

Additionally, recent studies commonly adopt transfer learning with conventional CNN architectures like VGG16, Inception v3, ResNet and related deep-learning pipelines, spanning apple and potato applications and even smartphone-ready vision-transformer systems [81–85]. Also, CNN-driven DL and transfer learning models used for cauliflower disease detection [86]. For apple leaves, object-detection and CNN pipelines have been reported, including an improved Faster R-CNN, an identification model, and robust CNN classifiers [87–89]. Other apple focused works explore specialized or transformer designs and segmentation using BAM-Net for challenging backgrounds, an improved ResNet-50 for disease and pest identification, deep transfer learning-based fire blight segmentation and vision transformers for accurate classification [90–94]. For tomato also an improved vision transformer was developed and a multi-scale attention on different feature types to classify disease was

used and in another work CyTrGAN used for artificial image synthesis also a ViT-RoT model for classify disease [95–98]. Beyond apple, tomato leaf diseases have been addressed with DL and explanation driven models, while potato lesions are segmented with an enhanced U-Net and soybean leaf diseases are classified using an optimized ConvNeXt architecture [99–102].

Besides, crops show analogous application in DL to detect infested soybean leaves, CNN-feature ensembles and Xception Network for grape, ConvNeXt-based and explainable models for mango and squeeze and excitation enhanced networks for strawberry disease detection [103–106]. Edge-computing oriented and lightweight approaches are emphasized in recent plant-disease work, a smartphone-based ViT system, coffee disease classification at the edge, a compact SK MobileNet for plant-disease recognition, and a lightweight YOLOv8 model for apple leaves [64, 81, 107–109]. Also, a compact CNN architecture is developed for classify mango leaf disease [110]. Other crop studies apply DL for disease identification in strawberry and tomato in parallel, smartphone-based and lightweight detectors illustrate deployment-focused directions [81, 101, 111]. Kalpana et al. was even able to classify diseases across multiple classes and was able to achieve an accuracy of 99.32% and [112, 113]. Recent DL and ML approaches for crop disease detection are compared in a Table 1 presenting models, accuracy, limitations and the direction of future work.

Table 1 highlights the recent ML and DL approaches applied to plant and crop disease classification. The results show that DL models such as CNN, EfficientNet, and YOLO variants consistently achieve higher accuracy compared to conventional ML approaches, particularly on well-curated datasets. However, most models are trained under controlled environments and face limitations in generalizing to real-field conditions. These findings highlight the need for more diverse datasets, lightweight architectures, and improved domain adaptation to enhance model robustness and practical applicability in precision agriculture.

Table 1 Summary of ML & DL related studies for plant and crop disease classification

Crop	Methodology	Key Models	Best Accuracy	Key Limitations	Future Directions	Reference
Tomato	DL	YOLOv8, EfficientNet, Hybrid CNN-Transformer	98.88%	Could not apply CycleGAN, controlled environment limitations	Integrate CycleGAN, enhance adaptability across diverse environments	[114–116]
Tea	DL	YOLOv5, Custom CNN, MobileNet	98.99%	Long training time, dataset restricted to a specific geographic region	Collect more diverse leaf images, integrate IoT and environmental sensor data	[61, 117, 118]
Coffee	DL	EfficientNet, MobileNet, Custom CNN	99.9%	Dataset includes many lab-captured images, real-field samples are limited	Use exclusively real-field images, include multi-disease classification	[63, 64, 66]
Grape	DL	VGG16, CNN variants, YOLOv8	98%	Performance drops on “Black Rot” class, generalization without augmentation is poor	Explore newer augmentation techniques, integrate state-of-the-art architectures	[52, 119]
Potato	DL	Custom CNN, MobileNet, DenseNet	99.2%	Model relies solely on image modality, only single disease per leaf considered	Extend to multi-disease recognition on a single leaf, incorporate multi-modal datasets	[56, 120]
Mango	DL	ConvNeXt, ViT, Custom CNN	99%	Model trained only on South India data, may not generalize globally	Apply to other crops, deploy in real-time disease detection systems	[121–123]
Cauliflower	DL	VGG19, YOLOv8, Custom CNN	99%	Dataset class imbalance, VGG19 model complexity vs. data size	Collect more diverse field images, use lighter models (e.g., MobileNet)	[75, 124]
Soybean	DL	YOLOv5s, VGG19, Improved ConvNeXt, DIM-U-net+LSTM	98.87%	Difficulty detecting overlapping leaves, limited to specific disease types	Expand dataset for overlapping leaves, extend to full-plant disease detection	[125, 126]

4.2 Hybrid ML–DL Approaches

Hybrid ML–DL approaches refer to frameworks that combine deep feature extraction with traditional ML classifiers, transfer learning, or few-shot learning strategies. Studies in this subsection are classified according to their architectural hybridization, independent of whether explainable AI or FL components are additionally employed, which are analyzed in later sections. Some works have studied hybrid schemes combining DL and classical, hand-crafted feature ML models or extending the pipeline of feature extractors through transfer learning. For example, in [92], the authors utilized a DL and ML hybrid pipeline approach where vegetation indices such as NDVI and GNDVI were derived from multispectral UAV imagery before being used to train an SVM model for disease identification in apple orchards [127]. Hybrid designs also had an impact on the detection of barley, apple and tomato diseases [128–132]. Across crops, methods include deep neural networks and a transformer-based fewshot pipeline for barley, alongside feature engineering, transfer learning and finetuning strategies reported for coffee and apple [133–137]. Both [134] and [133] leveraged a transformer-based DL backbone (Swin-B), under a few-shot learning (FSL) setting for the limited, in-house datasets. These methods showed impressive results and [134] recorded an accuracy of 91.80% on a 5-way 5-shot task. The application of DL–FSL hybrids reflects a general tendency toward implementing data-efficient approaches in precision agriculture. Swin Transformer is also used with EfficientNetV2 to capture local and global features and improve tomato leaf disease classification accuracy [138].

A two-branch architecture using CNN and Swin Transformer with a feature-fusion module to improve apple leaf disease classification in challenging real-world scenarios [139]. In tea crop analysis, the works shown in [118] and [140] employed hybrid models using DL-based feature evaluators and traditional ML classifiers [140] utilized EfficientNet-B0, ResNet-50 and DenseNet-121 models for deep feature extraction. Also, few short few-shot object detection model MAF-MixNet for tea disease detection [141]. Another work [118] generalized transfer learning (YOLOv7 with E-ELAN) to small tea leaf target detection in an unconstrained field environment, achieving near 89.1% F1-scores. Also, an integrated model combining YOLOv5 with attention-based mechanisms was developed for tea [142].

Two-stage or non-YOLO alternatives are still used when precise localization or task-specific constraints matter for example, TAME-Faster R-CNN for tea, a MobileNet-based online detection model for guava GLD-Det, a GAN-based approach for Verticillium wilt in Chinese cabbage DSCONV-GAN, and DIM-U-Net+LSTM for soybean [126, 143–145]. Additional recent work includes a YOLOv7-based tea detector, an improved lightweight YOLOv5 for strawberry, transfer learning-based DCNNs for strawberry classification [146–148]. A non-destructive grading model was developed for red grape external appearance using transfer learning and CNN–SVM fusion, especially under small sample condition [149]. Table 2 presents a detailed summary of hybrid techniques adopted for plant and crop disease classification, which highlights core algorithms, models, performance metrics, limitations, and possible future research directions.

4.3 XAI Approaches

This subsection focuses exclusively on explainability mechanisms applied to plant disease detection models. XAI techniques such as Grad-CAM, LIME, and SHAP are reviewed as model-agnostic or architecture-specific extensions that enhance transparency and trust, regardless of whether the underlying model is ML, DL, or hybrid in nature.

One of the most significant paradigms shifts in AI driven plant phenotyping is the transition from “black-box” models to interpretable, transparent systems through XAI. As DL models, such as Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs), demonstrate superior predictive accuracy on carefully-curated datasets, the need to understand how decisions are produced by these models is imperative for building farmer trust, enabling domain expert verification and ensuring robust deployment in agricultural environments [101, 150, 151]. XAI is now central to plant disease detection, offering both visual and quantitative insights into how

Table 2 Summary of hybrid related studies for plant and crop disease classification

Plant and Crop	Core Hybrid Architecture	Feature Source & Fusion Strategy	Classifier	Best Performance	Key Finding	Future Work	Ref
Apple	SVM+CNN	Vegetation indices (NDVI, ExB, etc.)	DL, ML	Precision: 92.8%, Recall: 91.2%	UAV-based system integrates multispectral analysis and SVM for accurate disease segmentation	Semi unsupervised labeling to reduce manual annotation	[92]
Barley	Transformer+Few-Shot Learning	RGB images+ Swin-B feature extraction	DL, Transformer	Accuracy: 91.8%	Combines transformer attention with low-data training strategy	Feature distillation for model compression	[134]
Barley	CNN+Transformer	Automated DL feature extraction with attention	DL, Transformer	Accuracy: 98.63%, F1: 98.60%	Superior performance with transformer-enhanced CNNs under small datasets	Extend dataset, automate annotation	[133]
Tea	YOLOv7+Transfer Learning	YOLOv7 enhanced with ELAN features	DL, Transfer Learning	F1-score: 89.1%, Precision: 92.2%	High precision in detecting small targets in open environments	Evaluate on multi-disease datasets, integrate GANs	[118]
Cauliflower	CNN+Transfer Learning	RGB pixel intensity histograms, morphological image features	DL, Transfer Learning	Accuracy: 99.90%	Advanced DL techniques for early disease prediction in cauliflower plants	Enhance contour feature extraction and extend system to detect diseases in other vegetables	[140]

models make decisions. Two main approaches lead the field: gradient-based methods like Grad-CAM and Grad-CAM++ and model-agnostic techniques such as perturbation-based explainers. These are often combined to provide complementary perspectives across different crops. An interpretable DL model leveraging Grad-CAM is proposed to visualize the infected leaf regions for mango [152]. A portable application, associating a lightweight wavelet-CNN with interpretability analysis in tea [60] and an interpretable study on Grad-CAM in end-to-end-based mobile app [153] enhances interpretability among diseases. In tomato, transformer-based detection [154] and CNNs with Grad-CAM integration [151] can help in the early and interpretable diagnosis. An interpretable DL model can enable accurate leaf disease classification in cabbage [155]. In strawberry, Grad-CAM explanation supports pest detection in smart-farm systems [156]. Lastly, a Grad-CAM enabled DenseNet-121 model generates distinct saliency maps for recognition of potato leaf disease [157]. These techniques allow to validate CNN-based models such as DenseNet121 [158] or Inception-based pipelines as the evidence that disease specific regions are indeed the source of classification. Furthermore, in tomato disease studies [151], Grad-CAM consistently identified necrotic or chlorotic areas that complemented the robustness of model predictions. Grad-CAM produces approximate localization maps that indicate which regions of an input image are highly informative for predicting the concept by using gradient information flowing into the final convolutional layer. For example, it was applied to visualize attention region of a Swin Transformer detector in identifying barley diseases [60] and present evidence for enhanced YOLOv8 model “TomatoDet” for tomato leaves [159].

Incorporation into real-time applications, e.g., the TeaNet8 Android App for tea leaf disease diagnosis [153], demonstrates its effectiveness in making the decisions of DL available to end-users. Mobile applications, such as TeaNet8 or PlantCare, are not only based on lightweight CNNs but also provide user-level interpretation using Grad-CAM. They can therefore readily deploy for on-site disease surveillance without the assistance of an expert. Model-agnostic explanations methods such as LIME (Local Interpretable Model-agnostic Explanations) and SHAP (SHapley Additive exPlanations), are widely used as well [36, 101, 117, 158, 160–162]. These approaches explain a particular prediction by approximating the complex model in the neighborhood of that prediction with an interpretable one. LIME has been critical for tools such as BotaniacX-AI [101], providing

fine-grained interpretability for post-hoc explanations. These methods are particularly useful in ensemble models or hybrid systems where the internal decision structure may be harder to decode. LIME was a building block of BotanicX-AI for tomato diseases where it provided pixel-wise explanation in post-hoc fashion. In addition, the XSE-TomatoNet model for tomato leaf classification combined LIME, SHAP and Grad-CAM to provide a multi-dimensional interpretability method [162]. This hybrid interpretability setup allowed the model to capture both spatial and feature-level explanations, which is crucial for fine-grained disease classification. The inclusion of SE-blocks in XSE-TomatoNet additionally aligned model attention with visual cues. In a multi-crop task, LIME was used to identify the pixels driving the predictions of an EfficientNetB0 model, increasing interpretability for end users (farmers) in combination with the PlantCare (mobile app) [161]. By highlighting region-specific cues for different plant species, LIME enhanced model generalization in varied datasets. This implementation helped users distinguish between disease symptoms across crop types.

Another interesting XAI method is the Saliency Maps [158, 163] which visualize the gradient of output score of any given class with respect to input pixels, and Score-CAM that is gradient-free but uses forward-pass score for each activation map as a weighting factor. In other work, PotatoGANs [164] compared Grad-CAM, Grad-CAM++ and Score-CAM to interpret the classifications by models such as DenseNet169 for a comparison of explanation fidelity in synthetic image-enhanced training. This comparison helped identify the most robust technique under varying image styles and data distributions. Grad-CAM++ frequently demonstrated superior performance in highlighting fine-grained disease lesions in GAN-synthesized scenarios.

XAI tenets are being integrated into the model by tailoring architectures through attention. Tomato, apple, tea and eggplant diseases study based on Squeeze-and-Excitation (SE) block, Convolutional Block Attention Modules (CBAM) and Transformer-based self-attention [64, 90, 118, 163, 165, 166]. These are mechanisms which enable the model to automatically recalibrate channel-wise and spatial feature responses, and their activation maps typically naturally highlight pathological regions, acting like a built-in explanation. For example, the BAM-Net for apple leaf disease adopted an Aggregate Coordinate Attention Mechanism (ACAM) to accentuate disease regions within complex backgrounds [90]. Table 3 summarizes the Explainable AI-based approaches reported in the literature for plant and crop disease classification.

Table 3 highlights the XAI approaches applied to plant and crop disease classification. Techniques such as Grad-CAM, LIME, and SHAP are widely integrated with CNN, EfficientNet, and Transformer models to visualize disease regions and improve interpretability. These methods enhance transparency and user trust, emphasizing XAI's essential role in developing reliable and explainable AI systems for precision agriculture.

Despite their widespread adoption, current XAI techniques in plant disease detection remain primarily qualitative and post-hoc, offering visual explanations without formal validation of causal faithfulness. Methods such as Grad-CAM, LIME, and SHAP highlight regions correlated with model predictions, but they do not guarantee that these regions correspond to true pathological causes. Moreover, most studies assess explainability visually, with limited involvement of agronomists or farmers to evaluate practical usefulness. These limitations indicate that while XAI enhances transparency and user trust, its reliability and real-world decision support value remain open to challenges requiring quantitative validation and user-centered evaluation.

4.4 FL Approaches

FL is examined as a decentralized training paradigm that operates independently of model architecture. Accordingly, this subsection reviews FL-based plant disease detection systems irrespective of whether they employ CNNs, EfficientNet, transformers, or hybrid backbones, with emphasis on privacy preservation, scalability, and non-IID data handling.

FL has emerged as a promising approach in agricultural technology to build robust ML models while keeping data privacy. Such a method is very suitable to plant disease detection since the data might be gathered from individual farms, research institutes and mobile platform [170]. There is an emerging literature that focuses on tuning the classic FL frameworks to address the specific characteristics of the agricultural domain. One of the key

Table 3 Summary of XAI based methods for plant and crop disease classification

Crop & Plant	Architecture	Model Used	Explainability Method	Dataset	Best Performance	Main Contribution	Future Work	Ref
Tomato	Transfer Learning	EfficientNet-B5	Grad-CAM, LIME	Kaggle (Public)	Accuracy: 99.07%	Explanation-driven DL framework	Explore alternative XAI techniques	[101]
Tomato	Object Detection (YOLO-based)	PLPNet (YOLOX-S baseline)	Attention Mechanisms (LRAM)	PlantVillage + Self-Collected	mAP50: 94.5%	Addressed intra-class variation & background	Develop disease grading systems	[165]
Apple	Instance Segmentation	Mask R-CNN (ResNet-50/101 + FPN)	Implicit (Segmentation Output)	Self-Collected	Precision: 92.8%	Segmentation for site-specific management	Explore transformer-based models	[92]
Apple	Lightweight CNN	ECA-DCMobileNet (based on MobileNet-V2)	Class Activation Mapping (CAM)	Self-Collected	Accuracy: 96.2%	Lightweight network with attention	Collect more varied environment data	[87]
Barley	Few-Shot Learning, Transformer	Swin Transformer, ViT, ResNet50	Grad-CAM	Self-Collected + Cassava Dataset	Accuracy: 91.80%	Few-shot learning for data scarcity	Reduce model complexity	[134]
Tea	Lightweight CNN	WaveLiteNet (MobileNetV3 + DWT)	CBAM Attention Module	Self-Collected	Accuracy: 98.70%	Lightweight wavelet-CNN model	Robust models for small occlusion	[60]
Tea	CNN with Attention	Res4net-CBAM	CBAM (Convolutional Block Attention Module)	Self-Collected	Accuracy: 98.27%	Automated detection in nature	Test multi-disease scenarios	[163]
Eggplant	Object Detection (YOLO-based)	EggplantDet (YOLOv8s)	Grad-CAM	Self-Collected	mAP: 92.61%	Lightweight model for environments	Adapt for adverse weather	[167]
Cabbage	Transfer Learning, CNN	EfficientNetB7, MobileNetV2, DenseNet201	XAI (Method not specified)	Self-Collected	Accuracy: 98.56%	DL system with XAI	Expand disease classes	[155]
Cauliflower	Transfer Learning, CNN	EfficientNet (B0–B4), Xception, etc	Grad-CAM, SHAP	Mendeley (Public)	Accuracy: 99.85%	Proposed GANs and XAI	Use lightweight hybrid models	[75]
Coffee	CNN	Multiple CNNs (AlexNet, VGG16, ResNet50)	Grad-CAM, Score-CAM	Multiple Public Datasets	Accuracy: 99.99%	Highlighted need for XAI	Combine DL with explainability	[64]
Strawberry	Transfer Learning, CNN	EfficientNet-B3, RegNetY-064	Grad-CAM	Proprietary + Open (AI Hub)	Accuracy: 85.6%	Pest classification with XAI	Apply to other crops	[156]
Potato	Transfer Learning, CNN	DenseNet-121	Grad-CAM, LIME	Kaggle (Public)	Accuracy: 98.37%	Hybrid model with XAI	Use drone-based capture	[157]

Table 3 (continued)

Crop & Plant	Architecture	Model Used	Explainability Method	Dataset	Best Performance	Main Contribution	Future Work	Ref
Mul-tiple Crops	Ensemble Learning	VGG16, VGG19, InceptionV3, ResNet101V2	LIME	PlantVillage (Public)	Accuracy: 0.923	Ensemble model with LIME	Integrate mul-tiple XAI techniques	[168]
Mul-tiple Crops	CNN & Transfer Learning	CNN (Architecture not specified)	LIME, Grad-CAM	New Bangladeshi Crop Disease Dataset (Public)	Accuracy: 89.75%	XAI for crop disease detection	Expand to more regions	[169]

reasons FL has significant promise in this area is due to the decentralized nature of agricultural data. A common argument found in the literature is to be able to train models on data that stays with their original owners or within local silos, farmers or specific research datasets, without having to share the data itself [155, 167–169]. Such a paradigm also guarantees data sovereignty which is particularly important for farmers and agriculture organization [171].

Advanced cryptographic methods have been additionally employed to achieve higher privacy. For instance, one approach integrated FedAvg with Differential Privacy (DP) and Homomorphic Encryption (HE), achieving 93.8% model accuracy on a dataset including PlantVillage and in-situ field images. This was at a reported 35% training overhead to a centralized model, which emphasizes the performance-privacy trade-off [172]. The popular client–server Federated Averaging (FedAvg) is adopted as predominant approach to aggregate the model in surveyed works [172–175].

Many DL models have been employed in the place of this training architecture. CNN is being widely used with the models such as VGG16, ResNet50, and DenseNet121 achieving superior accuracy beyond 98% in apple leaf disease, guava leaf disease detection task [170, 176, 177]. The growing use of MobileNet and EfficientNet in a number of studies indicates substantial shift towards lightweight and resource-friendly architectures. These architectures are well adapted for a mobile and edge computing context which is typically limited in terms of power consumption and bandwidth availability [173, 178]. Light models are prone to be deployed at edge. Models such as EfficientNet, for instance are frequently employed which can yield high accuracy up to 94.32% with specifications of Lightweight Federated Transfer Learning (LFTL) approach on EfficientNetB3 where the model size is limited at 4.32 MB [173]. In addition to CNNs, Vision Transformers (ViTs) had also been explored, which reported promising results, although its heavy computation cost is poses a limitation in FL [170].

A significant research focus has been on overcoming the challenges of statistical heterogeneity, or non-IID data, across clients. The standard FedAvg algorithm can experience performance degradation under such conditions and this led to more advanced aggregation methods being studied. Much recent work has centered on mitigating statistical heterogeneity, or non-IID data, among clients. Performance of the FedAvg algorithm can decays when such a situation occurs [175]. This has led to the study of more enhanced aggregation schemes. The AGRIFOLD model evaluated several algorithms such as FedProx, FedBN and SCAFFOLD and found that SCAFFOLD achieved the highest accuracy of 99.76% in non-IID environment [174, 175]. Other innovative methods using Wasserstein distance for regularization and devising Intelligent Weight Transfer (IWT) mechanism to optimize the aggregation for Federated DL. These FL systems are applied in a wide range of crops. For additional fruit crops, such as apple, guava, tomato and on staple crops like corn or rice [172–174].

Recent advancements in FL within agricultural applications reflect a growing interest in building decentralized, privacy-preserving systems for plant disease detection. Literature has explored different DL architectures, from traditional CNNs including VGG16 to ResNet50 and even more lightweight models such as EfficientNetB3, and Vision Transformers (ViTs) to improve model accuracy while keeping deployment efficiency. Several aggregation techniques other than FedAvg including FedProx, FedBN, SCAFFOLD and Intelligent Weight Transfer (IWT) are employed to handle non-IID data distribution and improve generalization.

Major novelties are utilization of cryptographic methods like differential privacy, homomorphic encryption, application of attention-based architectures (PMACNN) and fusion of multi-model data (image and environmental sensor). The robustness, scalability and real on farm application potential of these FL frameworks have been successfully explored in a variety of crop types including apple, guava, tomato, rice and maize. In summary, these contributions reflect the maturity and increasing level of sophistication of FL applications in agrotech bringing forward secure, intelligent and distributed plant health monitoring systems. Table 4 presents a summary of federated based approaches for plant and crop disease detection or classification with their feature and key contribution.

Table 4 presents recent applications of FL in plant and crop disease detection. The reviewed studies demonstrate that frameworks such as FedAvg, FedProx, and SCAFFOLD, combined with models like CNN, VGG16, and EfficientNet, attain strong performance while preserving data privacy. These findings emphasize FL's potential to enable decentralized, privacy-aware, and scalable AI solutions for agricultural diagnostics across diverse and distributed datasets.

Table 4 Summary of FL-based approaches for plant and crop disease classification

Dataset & Crop-Plant Focus	Model (Backbone)	FL Method & Aggregation Algorithm	Best Performance Metric	Key Features Used	Key Contribution	Ref
Private (4 classes), General plant leaves	EfficientNetB3, Dense Neural Network (DNN)	Client–Server FL, FedAvg	Accuracy: 94.32%	Features extracted by pre-trained models (EfficientNetB3)	Lightweight Federated Transfer Learning (LFTL) for edge devices	[173]
PlantVillage+field images (38 classes, 14 crops)	EfficientNet-B4	Client–Server FL, FedAvg+DP+HE	Accuracy: 93.8%, F1-Score: 93.3%	Leaf image features (visual patterns, texture, color)	Integration of Differential Privacy (DP) & Homomorphic Encryption (HE)	[172]
12 public datasets (10 classes, 16 crops)	VGG16 with Efficient Channel Attention (ECA)	FedAvg, FedProx, SCAFFOLD, FedBN, FedDF	Accuracy: 99.76% (SCAFFOLD)	Visual features from leaf images enhanced via ECA	AGRIFOLD framework with multi-dataset training and XAI	[174]
PlantVillage (54,305 images, 14 crops)	VGG16, DenseNet121, MobileNet, ResNet50	Horizontal FL, FedAvg (Flower framework)	Accuracy: 99.71% (VGG16)	Visual features (leaf textures, shapes, color patterns)	FL for scalability and cloud integration	[175]
7,832 images (5 apple disease classes)	Custom CNN (2 Conv Layers)	Federated Averaging (FedAvg)	Accuracy: 97–98%	Hierarchical visual features from RGB images	FL-based CNN for apple leaf disease recognition	[176]
Private (6 guava disease classes)	Convolutional Neural Network (CNN)	Federated Averaging (FedAvg)	Accuracy: 97–98%	Features extracted via CNN layers (conv+ReLU+pooling)	FL-CNN for guava leaf disease classification	[177]
GuavaLeaves8000 (5 diseases+healthy)	EfficientNet-B3	Federated Averaging (FedAvg)	Accuracy: 98.32%, AUC: 98.93%	Image-based features extracted by EfficientNet-B3	GuavaCareNet: Real-time diagnosis with edge-cloud integration	[178]
PlantVillage (50,000+ images of Apple, Grape, Corn, Tomato)	ResNet50, VGG16, ViT-B16, DenseNet121	FL (FedAvg)	Accuracy: 100% (ResNet50 on Grape), 98.57% (ViT-B16)	Raw RGB leaf images, image patches for ViTs	Comparative analysis of CNNs and Vision Transformers (ViTs) in FL	[170]
Private data from Indian farmland	CNN, RNN, LSTM, GRU	FL (Client–Server)	Accuracy: 0.99 (CNN)	Sensor data (environmental), Image data (leaf health)	FL with IoT sensor and image data for sustainability	[171]
PlantVillage (46,672 images, Apple, Cherry, Maize, Tomato)	H-CNN	Federated DL (FDL), FedAvg, FedAdam	Accuracy: Up to 98%	Fused low- and high-level features via bi-stage convolution	Federated DL (FDL) with lightweight H-CNN	[179]
PlantVillage, Kaggle, Mendeley (90,516 images, 23 classes)	PMACNN, InceptionV3	Federated DL (FDL), Intelligent Weight Transfer (IWT)	Validation Accuracy: up to 99%	Leaf image features extracted via PMACNN (attention-based)	Adaptive knowledge transfer for non-ID data (FDL-IWT)	[180]

The performance metrics reported in Table 4 indicate that FL models achieve results that are largely consistent with those obtained by centralized ML, DL, and hybrid approaches summarized in the preceding performance tables. In studies where comparable datasets and backbone architectures are used, federated models generally exhibit similar accuracy and F1-score ranges, suggesting that decentralization does not inherently lead to substantial performance loss. Minor degradations are primarily observed in scenarios with strongly non-IID client distributions, whereas more balanced partitions yield performance closely matching centralized counterparts. These observations support the view that FL can preserve competitive predictive capability while enabling distributed and privacy-aware model training, although conclusions remain contingent on dataset characteristics and evaluation protocols.

Table 5 demonstrate that decentralized learning methods (especially FL) could attain the performance close to centralized ML/DL techniques. While FL-based models have marginally lower performance than their centralized counterparts (eg, 93–98% on decentralized ones versus 97–99% with centralized training), this reduction is attributed to the distributed setting of the learning process. In federated environments, models are trained in a decentralized manner at clients over heterogeneous and non-IID data. Therefore, the global model voting is probably not able to understand the statistical properties of all data, and thus marginal differences in convergence and accuracy occur. Furthermore, communication limitation, local computation constraints and asynchronous model updates can also cause minor misalignment on parameter estimation, leading to other performance effects. However, the compromise is rational as FL's gains in ensuring privacy of data and effectiveness of communication are substantial. With FL allowing for shared model training without sharing raw data, the threat to privacy is significantly mitigated and the compliance of data protection standards can be guaranteed. In general, we confirm that model accuracy can degrade slightly for federated models, but on the other hand offering a favorable balance between model performance and reduced communication.

While most reviewed studies do not evaluate centralized and FL under fully identical experimental settings, several works employ the same datasets, model architectures, and evaluation metrics to enable approximate comparison. As summarized in Table 5, FL models typically achieve performance close to centralized counterparts, with modest accuracy reductions attributable to decentralized optimization, non-IID data distribution, and communication constraints. These findings suggest that, under comparable assumptions, FL offers a favorable trade-off between predictive performance and data privacy rather than a direct performance replacement for centralized training.

It should be noted that performance values reported in Tables 1–5 originate from different datasets and evaluation setups and therefore should not be interpreted as a unified ranking of methods. Instead, these tables provide an aggregated overview of reported outcomes to highlight general performance tendencies and methodological directions in the literature.

Figure 3 illustrates the comparative accuracy between centralized ML/DL and decentralized FL frameworks. Even with minor performance differences, FL models demonstrate competitive accuracy while offering significant benefits in data security and scalability, reinforcing their value for privacy-preserving collaborative learning in precision agriculture.

Despite the advantages of FL for privacy-preserving collaboration, several practical challenges remain. Reviewed studies report increased communication overhead due to repeated client–server model exchanges,

Table 5 Comparison of ML/DL (centralized) and FL (decentralized) in terms of accuracy, data privacy, and communication efficiency

Approaches Used	ML/DL Centralized	FL Decentralized	Data Privacy	Reduced Communication Cost	Ref
FedAvg+DP+HE	97.2%	93.8%	✓	✓	[172]
FL+VGG16+XAI	99.9%	98.8%	✓	✓	[174]
VGG16+FL	99.80%	99.71%	✓	✓	[175]
CNN+FL	99.2%	98%	✓	✓	[177]
CNN+FL	99%	94%	✓	✓	[171]
H-CNN+FL	98%	93%	✓	✓	[179]

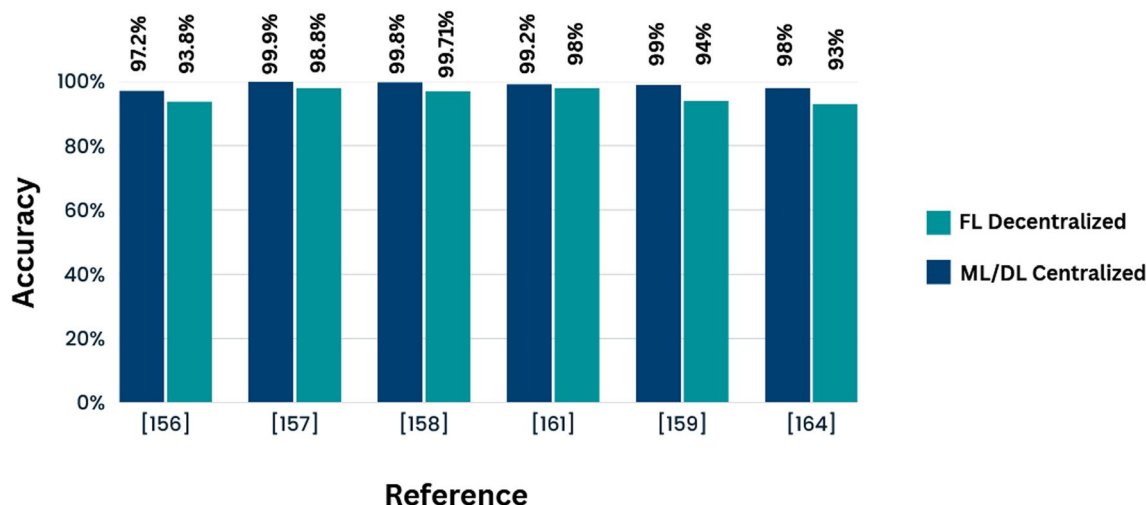


Fig. 3 Accuracy Comparison Between Centralized ML/DL Models and Decentralized (FL) Methods

particularly when deep architectures and large client populations are involved. Convergence instability is also observed under non-IID data distributions, where standard aggregation methods such as FedAvg may lead to slower or suboptimal convergence. In addition, scalability constraints arise from heterogeneous client capabilities, limited bandwidth, and asynchronous participation. While advanced aggregation strategies and lightweight models partially mitigate these issues, systematic evaluation of communication efficiency and large-scale deployment remains limited.

5 Results and Analysis

The performance results summarized in this section are derived from heterogeneous experimental settings, including variations in dataset size, class balance, crop type, imaging modality, and evaluation protocols. As a result, reported accuracy metrics are not treated as directly comparable across studies. Instead, they are analyzed alongside contextual factors such as dataset origin (public vs. self-collected), environmental conditions (controlled vs. real-field), task formulation (classification vs. detection), and model architecture. Where alternative metrics such as F1-score, precision, recall, or mAP are reported, they are highlighted alongside accuracy to better reflect performance under class imbalance. This approach enables a structured synthesis of methodological trends and trade-offs while avoiding misleading cross-dataset comparisons.

This section provides a comprehensive interpretation of the findings derived from the systematic review of 145 peer-reviewed studies focused on the application of plant disease detection. Figure 4 shows the temporal change of the number of publications for the period 2023–2025 in terms of the methodologies. Interestingly, the trend for the use of DL and explainable AI methodologies has been increasing over time, demonstrating that people are now paying more attention to performance accuracy and model interpretability. Moreover, the increasing use of FL models in recent articles additionally highlights a shift towards privacy preserving frameworks that are crucial for decentralized agricultural systems.

As illustrated in Fig. 5, the largest share of literature (30%) came from others (Frontiers, arXiv, ACS, QASCF and more) followed closely by Elsevier (19%) and MDPI (19%). IEEE Xplore and Nature accounted for 12% and 6%, respectively, of the corpus, with Springer at 6%. This distribution underscores the multidisciplinary character of the field, as significant contributions cross computer science, large scientific repositories and high-impact, multidisciplinary journals.

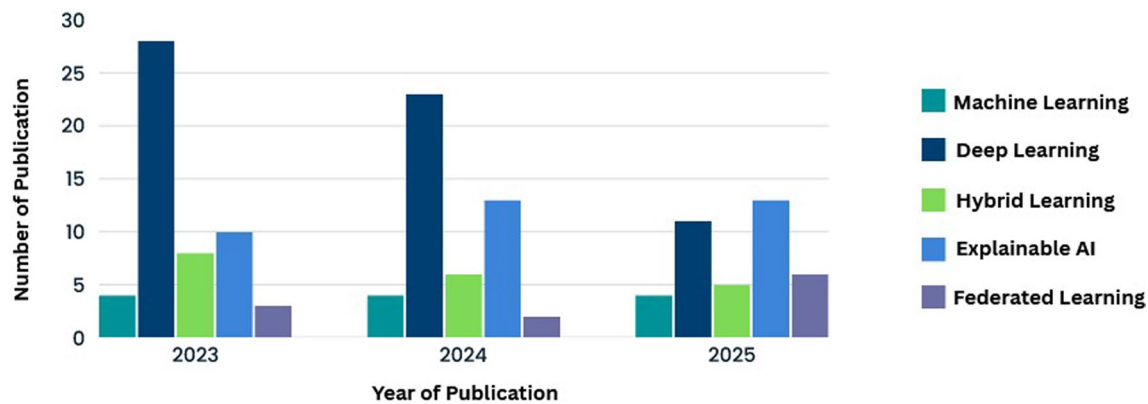


Fig. 4 Publication Trends of plant and crop disease classification

Fig. 5 Percentages of articles published in main journal

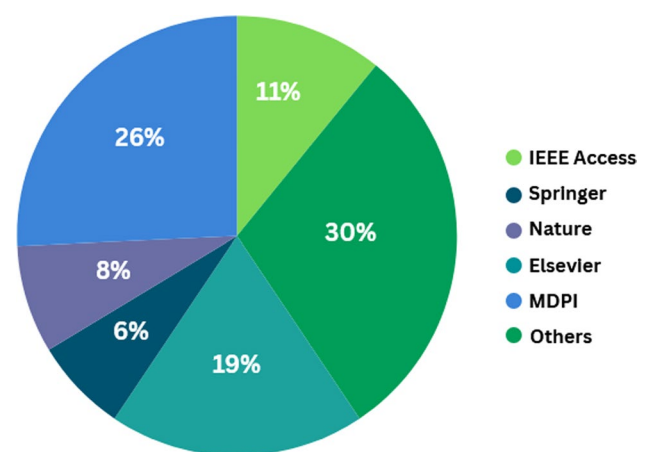


Table 6 Overview of plant and crop disease detection application

Application Name	Deployment Type	Model	Notable Features	Ref
TeaNet8	Android	ResNet50V2 + Grad-CAM	Real-time leaf diagnosis, XAI-based visualization	[153]
GFDI System	Android	DenseNet169	Disease detection + pesticide recommendation	[181]
PCD App	Android (Offline)	MobileNetV2	Works offline on low-end devices, transfer learning	[182]
PlantCare	Android	EfficientNetB0 + LIME	Multi-crop support, transparency, multilingual UI	[183]
Guava Web App	Web (Flask)	CNN (AlexNet-Inception)	Small architecture, real-time online use	[55]
Cabbage Web App	Web (Flask)	EfficientNetB7	Real-field data, high accuracy, limited classes	[155]

5.1 Real-World Applications of AI in Plant Disease Detection

Although many promising plant disease detection models being proposed using ML and DL, their integration into real-time field implementable systems is still limited. To fill this gap, recently, there has been an increasing number of studies that have aimed at the development of mobile and web-based systems, which can do on-device inference with supporting interpretability through XAI while providing an intuitive interface. Few such applications have been compiled in Table 6.

Regarding Android-based, based mobile applications, an app named TeaNet8 [153] was designed for 8 types of tea leaf diseases detection by employing a fine-tuned ResNet50V2 model along with Grad-CAM to visualize important lesion zones which could work perfectly on device with accuracy reaching 97%. In the same way, system to target GFDI [181] mango leaf disease classification and pesticide recommendation. Built using React

Native and Node.js, has been tested with 43,057 images of 20 types of trash, DenseNet169 achieves an accuracy of 97.81% and the time consumed in real-time detection for each image is about two seconds. PCD App (Potato Crop Diseases) is designed to work offline on low-end Android devices with MobileNetV2, and capable of detecting three disease classes by Tensor Flow Lite deployed lightweight (77 MB) models that are well-suited for rural farming areas [182]. Another platform, PlantCare [36], which combines EfficientNetB0 with LIME to deliver explainable predictions covering 14 plant species and 38 disease classes, featuring multilingual support, transparent model outputs, post-inference analytics using Power BI.

Concurrently, there has been an effort to institute web-based applications as a solution to eliminate installation needs and improve accessibility. For instance, a web application for guava disease detection based on Flask utilizes a small AlexNet-Inception hybrid CNN trained with minimal data to achieve 93% accuracy with small model size for quick online inference [55].

Further, cabbage leaf disease detection, proposed system based on EfficientNetB7, DenseNet201, and MobileNetV2 was deployed through a local Flask server and reach an accuracy rate up to 98.56% for three classes of the disease (aphid, worm cuts and black rot) on real-field images. Typically, these are backed by Flask, Django or Node.js backend for universal access and centralized updating, representing a milestone towards applied AI integration in precision agriculture [155].

6 Discussion

6.1 General Insights and Trends

This review highlights a significant transition in precision agriculture from manual observation to intelligent and data-driven systems for plant and crop disease detection. ML and DL models have shown remarkable success in identifying diseases quickly and accurately. Architectures including CNNs, EfficientNets, and Vision Transformers often achieved higher than 95 percent accuracy, clearly outperforming traditional methods. Yet many of these models still struggle to maintain performance under real farming conditions where lighting, background, and weather vary widely. This gap shows that improving environmental robustness is still a key research priority.

A noticeable trend in contemporary research is the move toward building systems that are both explainable and privacy-aware. Explainable AI tools like Grad-CAM, LIME, and SHAP now help visualize how models make decisions. These explanations make it easier for farmers and agronomists to trust the technology and to verify that the models are focusing on real symptoms rather than background noise. At the same time, FL is becoming a powerful solution for privacy-preserving collaboration. Since agricultural data are often scattered across farms and institutions, FL allows models to learn from distributed sources without sharing sensitive data. Frameworks such as FedAvg, FedProx, and SCAFFOLD have achieved strong results, proving that privacy and accuracy can coexist in agricultural AI. Another notable trend is the increase in hybrid and compact models that balance accuracy, interpretability, and deployment efficiency. These models make it possible to bring disease detection to mobile devices, drones, and low-cost edge systems, enhancing practical applicability.

A joint analysis of Tables 1–5 reveals several consistent trends across methodologies. DL models generally achieve higher accuracy on controlled datasets, while hybrid and lightweight architectures offer improved robustness and deployability in resource-constrained settings. Explainable AI techniques, summarized in Table 3, are predominantly applied to CNN-based models and enhance transparency without significantly affecting predictive performance. FL approaches in Table 4 demonstrate accuracy comparable to centralized training (Table 5), albeit with modest performance trade-offs driven by communication and non-IID data challenges. Collectively, these tables highlight a trade-off landscape among accuracy, interpretability, privacy, and scalability, underscoring the need for integrated explainable–federated frameworks for real-world precision agriculture.

Despite these developments, challenges remain. Most datasets are limited in diversity and collected under controlled conditions, making it hard for models to generalize. Future progress will depend on richer datasets,

standard evaluation benchmarks, and a closer union between explainable and federated AI. Together, these approaches can create transparent, reliable, and scalable systems to support sustainable and resilient agricultural practices.

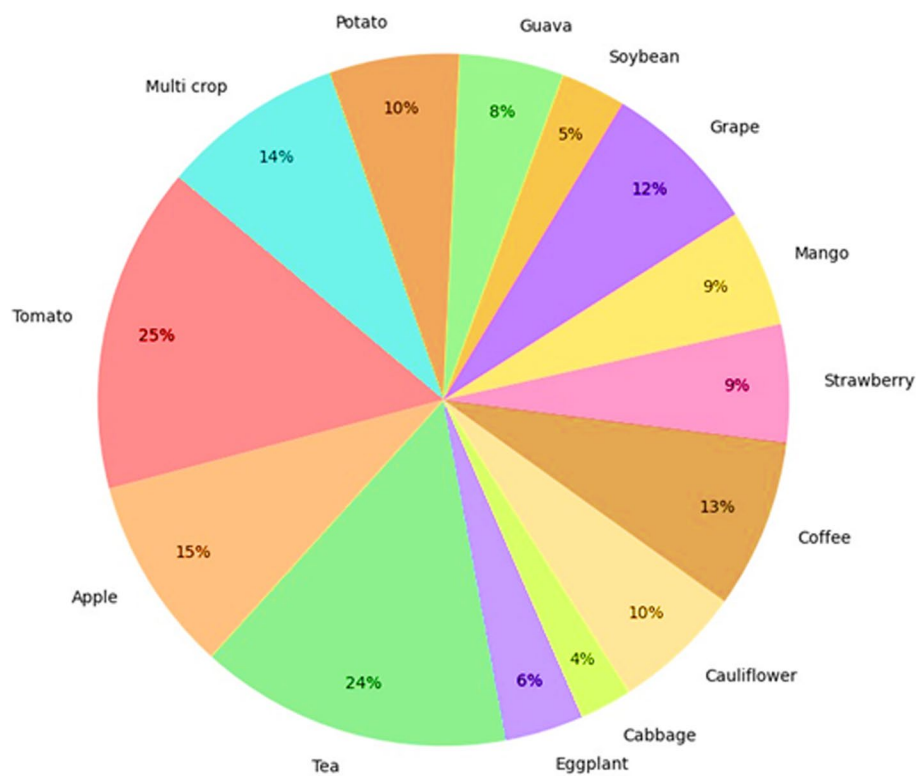
6.2 Discussion Based on Research Questions

6.2.1 RQ1 (Plant Species, Diseases and Dataset Generalization) Related Discussion

Analysis of recent DL-based plant and crop disease detection studies reveals a marked concentration on a limited number of economically significant crops. Figure 6 shows that tomato (25%) tea (24%) and apple (15%) are the top three species in current research emphasis which then is followed by multi-crop systems (14%), coffee (13%) and grape vines (12%).

Tomato has also been the subject of most crop models, leading to the development of numerous methodologies with a specific focus on important foliar diseases such as Early Blight, Late Blight, Leaf Mold, Septoria Leaf Spot and Tomato Yellow Leaf Curl Virus utilizing techniques like CNNs and object detection frameworks: YOLOv5 and YOLOv8. There's often Apple-Scab, Cedar-Apple-Rots, Black-Rot on studies focused on the genus *Sorbus* and Blister Blight and Anthracnose is what you look at when working with Camellias. Crops such as potato, cauliflower, strawberry and guava are other less predominant crops that also appear in disease detection literature. This distribution highlights a heavy research bias toward high-value horticultural species, often motivated by the presence of rich public datasets like PlantVillage and Kaggle and well-defined disease classes. Most reviewed studies utilized public datasets that contain thousands of images across multiple disease classes, with accessible metadata regarding disease type and severity. The PlantVillage dataset, which has been predominantly applied for research purposes, makes available a large number of pre-labelled images from many plant species and disease classes. Other commonly used public repositories are Kaggle, Mendeley Data, Roboflow Universe and AI Challenger dataset. These public datasets usually consist of a large number of images. However, an important proportion of

Fig. 6 Plants and crops used in disease classification studies



studies also use private or self-collected datasets, curated usually to tackle particular regional disease outbreaks. A key problem is that these private datasets are rarely released, which makes it difficult to independently validate, and for the wider community to benchmark and generalize models. Thus, while data availability for first order model development is high for some major crops through the provision of public datasets, the lack of open access to a wide range of diverse and specialist private collections has remained a key bottleneck against developing fully comprehensive and generalizable detection systems. Despite of availability of large public datasets, the limited accessibility of diverse private collections continues to hinder the development of robust and generalizable plant disease detection systems.

Table 7 summarizes the plant disease detection data sets and summarizes them for comparison covering crop types, sources of database, number of images in each one, the number of classes, accessibility to other users taking into account subject title references. It reflects that while there are many widely researched crops such as tomato, apple and tea which possess quality public datasets, the likely of high usage and the number commonly cultivated crops from private collections is significantly high like barley, eggplant and cabbage. This leads to an unequal distribution of available data in the various crops.

Table 7 highlights that most studies rely on public datasets such as PlantVillage, Kaggle, and Mendeley, while many region-specific collections remain private. This uneven data availability limits model generalization and comparability, emphasizing the need for more diverse and open-access field-based datasets.

6.2.2 RQ2 (Environmental Impact on Model Robustness) Related Discussion.

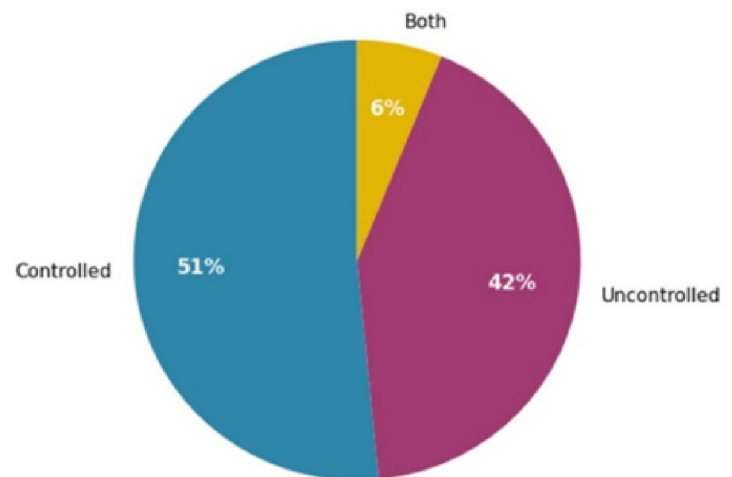
The plant and crop disease detection models examined are mostly created and evaluated on datasets captured under controlled growing conditions, namely in a uniform and well-controlled setup where images are taken with diseased leaves laid under plain or white backgrounds, in consistent lighting conditions, and free of environmental noise such as shadows, occlusions or background noise. These conditions aid in making feature extraction by the model easier and decrease intra-class variability, which results in very high reported accuracies. For instance, models such as YOLOv5, YOLOv8 and transformer-based networks are found to perform well when applied on curated datasets like PlantVillage or Kaggle. On the other hand, uncontrolled or real-field settings are significantly challenging to computer vision systems. Field acquired images display different sunlight conditions, may contain shadows and overlapping leaves, are with soil or other plants as background and are taken from disease-affected regions in various poses or with occlusions. Introducing variability in the form of these factors is even harder for disease detection. Models that are trained on controlled data sets tend to over-fit towards perfect

Table 7 Overview of major plant and crop disease datasets emphasizing crop coverage, number of images, disease classes, and accessibility level

Plant/Crop	Dataset	Number of Images	Number of Classes	Accessibility	Ref
Tomato	Roboflow Universe	3000	4	Public	[115]
Apple	PlantVillage	3171	4	Public	[85]
Barley	Self-Collected	2165	5	Private	[134]
Tea	Mendeley	2824	8	Public	[153]
EggPlant	Self-Collected	4930	5	Private	[167]
Cabbage	Self-Collected	3296	4	Private	[184]
Cauliflower	Mendeley	7402	4	Public	[75]
Coffee	The BRACOL and DPCL datasets	1040	4	Public	[65]
Strawberry	Self-collected	1263	4	Public	[54]
Mango	MangoLeafBD	4000	8	Public	[122]
Grape	Grape Leaf Dystrophy Dataset	5500	3	Public	[183]
Soybean	Mendeley	6410	3	Public	[185]
Guava	Guava Patches Dataset (GPD) and Guava Leaf Diseases Dataset (GLDD)	3738	5	Public	[186]
Potato	Plantvillage	2152	3	Public	[57]

Table 8 Overview of environmental conditions, imaging devices, and regional sources associated with datasets used in plant and crop disease detection research

Plant/Crop	Dataset capturing condition	Device Used	Region of Data Collection	Dataset Source	Ref
Apple	Uncontrolled Environment	Logitech C920 webcam	Pennsylvania, USA	Self-Collected	[92]
Tomato	Controlled Environment	SONY DSC-HX400 digital camera	Guangdong, China	Self-Collected	[151]
Tea	Controlled Environment	Canon EOS 250 d Camera	Sylhet, Bangladesh,	Self-Collected	[187]
Strawberry	Uncontrolled Environment	SVC3 camera	Tiptree, UK	PlantVillage	[103]
Soybean	Both	Sony DSLR-A350	Madhya Pradesh, India	Self-Collected	[188]
Guava	Controlled Environment	5 MP RGB Raspberry Pi Camera	Durgapur, India	Self-Collected	[45]
Cabbage	Uncontrolled Environment	Specim FX10 hyperspectral camera	Korea	Self-Collected	[46]
Eggplant	Controlled Environment	Multispectral camera FS-3200 T-10GE-NNC	China	Self-Collected	[77]

Fig. 7 Percentage distribution of controlled, uncontrolled, and mixed environments across plant disease detection studies

conditions and do not generalize well when applied in practice, where these kinds of variables are not controlled for. The reviewed papers in general recognize such limitations, some of them clearly mentioning that their models have not been validated in-field while others highlight it as future work. There was also one paper mentioning that challenges when using methods such as CycleGAN for domain adaptation are the lack of real-world data. Hence, an overview of related works combining the environmental conditions, imaging devices and regional sources has been showed in Table 8.

Table 8 shows that dataset collection conditions vary widely across studies, ranging from controlled laboratory setups to real-field environments using different imaging devices and regions. Such variation strongly influences model robustness and highlights the need for standardized, environmentally diverse data acquisition practices.

Figure 7 demonstrates that 51% of models were trained on controlled data, 42% on field datasets, and only 6% on both indicating very few models are trained to handle mixed environmental conditions. This disparity explicitly underlines a lack in training models to address mixed or varied environmental conditions, more representative of the real-world agricultural conditions. Environmental information (date, imaging system, and location) during data collection process for the cropped plots can be found in the supplementary table separated by crop type. Such metadata is useful for evaluating of how their performance can be robust and transferable across different farming areas.

6.2.3 RQ3 (Model Types, Performance and Scalability) Related Discussion

Various ML, DL and hybrid ML–DL models were explored for plant disease detection with varying degree of effectiveness in terms of accuracy, efficiency, and scalability. As summarized in Table 1, traditional ML models such as SVM, Random Forest and ANN obtained accuracies from 82 to 98%, with high interpretation/low

computational burden but low generalization under real-world diversity. DL methods such as CNNs, ResNet, EfficientNet, YOLO variants, often achieving high accuracy often above 99%.

Hybrid ML–DL models shown in Table 2 integrate deep feature extraction with traditional ML classifiers or transfer learning frameworks, reaching balanced accuracy with enhanced efficiency for limited and diverse datasets. These hybrid models improve robustness and flexibility with moderate interpretability, which is of great potential in actual agricultural application. Detailed comparison of these techniques and their future trend observed in Table 1 and Table 2. In general, the ML models are efficient but fall short of scalability, DL models exhibit high accuracy at cost of computational resources where hybrid frameworks offer a balanced trade-off between the accuracy, generalization and interpretability.

6.2.4 RQ4 (Role of XAI and FL) Related Discussion

XAI and FL are emerging key topics to increase the transparency, trust, and privacy of data in AI-driven plant disease detection systems summarized in Table 3. The XAI methods like Grad-CAM, LIME and SHAP have been commonly combined with CNN, EfficientNet and Transformer based models to visualize model attention in order to emphasize affected leaf areas on plant leaves. Such approaches enhance interpretability by allowing experts or farmers to “understand” what the model is predicting and are able to reach accuracies of over 97–99% while preserving interpretable visual reliability and transparency. Furthermore, attention mechanisms such as the CBAM and self-attention modules inherently boost model interpretability in that they concentrate on task-relevant regions of leaves, which facilitate real-world application interpretability.

As summarized in Table 4, FL frameworks aim to achieve privacy-preserving collaboration over decentralized agricultural data, addressing the growing demand for secure and decentralized model training in precision agriculture. FL methods like FedAvg, FedProx, and SCAFFOLD allow learning on local data without sharing raw images and respecting the human’s data privacy and sovereignty. Hybrid studies: Studies that combines both CNN and EfficientNet in addition to Vision Transformers under the federated setting in which they achieved accuracies under 99% while using methods such as Differential Privacy (DP) and Homomorphic Encryption (HE) for improving on privacy during aggregation. These frameworks scale effectively through multiple farms and institutions with strong generalization performance under non-IID (heterogeneous) data settings.

In summary, Tables 3 and 4 indicate that XAI enhances model explainability and transparency, while FL ensures data privacy and collaborative scalability. Together, these approaches represent key advancements toward developing reliable, interpretable, and privacy-aware AI systems for sustainable and intelligent plant disease management.

6.2.5 RQ5 (Challenges & Future Directions in AI for Plant Health) Related Discussion

Future advancements in the development of scalable, interpretable, and environmentally resilient artificial intelligence (AI) systems for detecting plant diseases must go beyond traditional objectives of attaining high accuracy in controlled laboratory settings. Future smart farming systems should focus on being strong, easy to understand, and able to grow in different types of environments with different types of light, plants, and background changes.

A major challenge in current ML-based plant disease models is that they struggle to generalize well to open-field conditions. Model performance is commonly reduced by variation in environmental predictors including light intensity, soil reflectance, humidity and leaf occlusion. To address these issues, domain adaptation (DA) and data augmentation methods are extensively studied to improve environmental robustness that allow models learning invariant representations across domains. Furthermore, incorporation of hybrid architectures like CNN Transformer fusion networks could combine the local feature extraction capability of convolutional layers with global context modeling power of attention mechanisms to design accurate and portable architectures across agro-ecological backgrounds.

Scalability is a key area for enhancement. Real-life applicability for deployment purposes in the field, especially in low-resource rural agricultural settings requires model to be lightweight and computationally efficient enough to run on edge devices like mobile phone, drones or IoT enabled sensors. Methods such as model pruning, quantization, and knowledge distillation can reduce the base model size without sacrificing performance. FL offers scalable, privacy-preserving training. Based on decentralized collaboration learning, FL allows multiple farms or institutions to cooperatively update a mutual model without exchanging raw data, which can protect the self-control of data, and it also improve global performance by diversity across local datasets.

Also important is the integration of XAI methods for improve model transparency and interpretability, essential for building trust among agronomists, farmers, and decision-makers. By analyzing which visual or spectral features are most relevant to making a model's prediction, efforts can be applied for both scientific validation and practical ease of use techniques such as Grad-CAM, LIME and SHAP. These explanations not only enhance end-users' trust, but also give diagnosing explanation of bias for model and misclassification.

Future research directions can be directly aligned with identified limitations: poor real-field generalization calls for domain adaptation and mixed-environment datasets; data imbalance motivates few-shot and synthetic data generation; deployment constraints require lightweight and compressed models; privacy concerns favor communication-efficient FL; and limited trust in predictions highlights the need for quantitatively validated and user-centered XAI methods.

6.3 Environmental Robustness Challenges in Real-Field Deployment

Although many AI-based plant disease detection models achieve high accuracy under controlled laboratory conditions, their performance often deteriorates in real-field environments. The reviewed literature reveals several recurring factors that limit environmental robustness. Variations in illumination, shadows, soil textures, and complex backgrounds introduce visual noise that reduces feature discriminability, particularly for models trained on uniform datasets. In addition, real-world scenarios frequently involve leaf overlap, partial occlusion, and multi-scale disease symptoms, which challenge reliable lesion localization and detection. Dataset bias further exacerbates this issue, as many studies overrepresent clear or early-stage disease symptoms while underrepresenting mild, mixed, or late-stage infections, leading to reduced generalization. Moreover, domain shifts caused by differences in crop varieties, geographic regions, imaging devices, and acquisition protocols significantly affect model transferability across farms and institutions. While recent studies explore mitigation strategies such as data augmentation, domain adaptation, and multi-environment training, large-scale real-field validation remains limited, underscoring the need for standardized benchmarks and mixed-environment datasets to bridge the gap between laboratory performance and practical deployment.

7 Conclusions

This systematic review provides a comprehensive synthesis of latest progress in ML, DL, XAI, and FL for plant and crop disease classification within precision agriculture. The collective evidence from 145 peer-reviewed studies reveals a steady paradigm shift from conventional manual inspection to automated, intelligent, and data-driven systems enabling real-time disease detection. Deep architectures particularly CNNs, EfficientNets, and transformer-based models demonstrate superior diagnostic accuracy on curated datasets, yet their robustness under uncontrolled environmental conditions remains a persistent challenge. Hybrid ML–DL frameworks have shown promise in balancing accuracy, efficiency, and interpretability for limited or heterogeneous data scenarios.

The adoption of XAI methods such as Grad-CAM, LIME, and SHAP marks a crucial advancement toward transparent AI systems that bridge the gap between computational decisions and human trust. These methods help validate models, bias identification, and practical visualization for agronomists and farmers. Concurrently, the emergence of FL frameworks introduces decentralized intelligence and privacy-preserving collaboration across

geographically distributed agricultural data sources. Algorithms like FedAvg, FedProx, and SCAFFOLD have demonstrated high performance even under non-IID data conditions, offering a compelling trade-off among accuracy, privacy, and communication efficiency.

Future research should focus on combining XAI and FL into scalable, energy-efficient architectures that can operate on edge or mobile platforms. Addressing dataset imbalance, cross-domain generalization, and environmental adaptability through domain adaptation, data augmentation, and lightweight model compression will be critical to achieving real-world deployment. Furthermore, integrating multimodal data streams from hyperspectral imaging to IoT sensor networks can enrich contextual learning and enhance early disease prediction.

Ultimately, the fusion of federated and explainable AI represents a transformative pathway toward resilient, interpretable, and sustainable plant disease detection systems. Such developments will empower next-generation precision agriculture, maintaining transparency, privacy, and global food security in an era increasingly defined by data-driven decision-making.

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